

# Dynamic Pricing for European Camping Group

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# Presentation Overview

## 1 The Problem

250K prices to set every year, one pricing trap

## 2 The Fix

Causal identification via Bookings on Books (BoB)

## 3 Two Models, Two Jobs

Bayesian (estimate) + LightGBM (predict & optimise)

## 4 What the Optimiser Recommends

Bidirectional price changes + €63M projected uplift

## 5 Next Steps

Randomised experiment + pipeline with dashboard

# The Problem

**400+**

Campsites  
across Europe

**10**

Countries  
in operation

**250K+**

Prices to set  
every year

*ROMGID = campsite  $\times$  accommodation  $\times$  market  $\times$  arrival week*

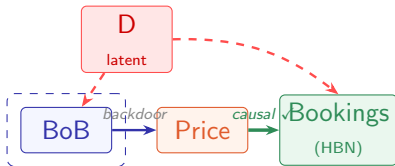
**Can historical booking data recommend better prices and increase ECG's revenue?**

# The Fix: Causal Identification

Naïve model (wrong):

$$\hat{\beta} = +0.29$$

"Raising prices  
= more bookings?"



After conditioning on BoB:

$$\hat{\beta} = -0.76 \checkmark$$

Inelastic:  $|\beta| < 1$   
revenue rises

ECG's pricer reacts to demand: high demand  $\rightarrow$  prices  $\uparrow$ , low demand  $\rightarrow$  prices  $\downarrow$ . High prices & high bookings co-occur in the data  $\Rightarrow$  naïve regression gets the wrong sign. Conditioning on BoB closes this backdoor.

# Two Models, Two Jobs

## Model v3 · Bayesian

*Goal: estimate elasticity*

$$\hat{\beta} = -0.76$$

*inelastic demand,  $|\beta| < 1$*

- Minimal: only BoB + price
- Cannot make future predictions
- Revenue rises when price rises on average

## LightGBM · Gradient Boosting

*Goal: predict & optimise*

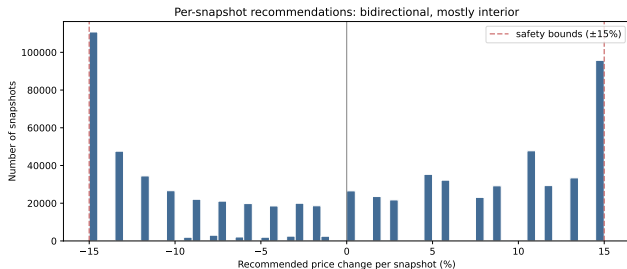
$$\hat{\beta} \approx -0.75 \checkmark$$

*confirms Bayesian estimate*

- Adds lead time & calendar features
- Out-of-sample error: 27% per ROMGID
- Flexible demand curve → interior optima

# What the Optimiser Recommends

Recommended price changes (all snapshots)



**72.6%**

interior optimum (not at boundary)

**€63M**

predicted uplift (+38.8%)

**51/49**

green / yellow flag split

*The €63M uplift is a model projection and likely optimistic. A randomised experiment is needed.*

# What ECG Should Do Next

## 1 Randomised price experiment

100–200 ROMGIDs · prices randomised  $\pm 25\%$  · one 30-week season

⇒ unbiased elasticity estimate · tests beyond  $\pm 15\%$  safety range

## 2 Pipeline + dashboard (nightly rerun)

Recommendations.csv per ROMGID · green/yellow flags · sortable dashboard

⇒ builds managerial trust · managers audit anomalies, not all prices

*Both tracks run in parallel*

# Conclusion

$$\hat{\beta} = -0.76$$

True price elasticity  
after closing  
the backdoor

**72.6%**

Snapshots at interior  
optimum, bidirectional

**2 tracks**

Experiment + pipeline  
to verify & deploy

**ECG is leaving money on the table.**

*A small randomised experiment + a validation  
pipeline will answer how much, at minimal cost.*